

CLEREVA Student Success Assistant — Final Report

Identifying Academic Risk in Portuguese Secondary Schools

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1 Summary

This report presents a reproducible data analysis of academic risk in Portuguese secondary school Mathematics students ($N = 395$). Using the UCI Student Performance dataset, we applied six statistical skills to (1) describe performance patterns, (2) compare student groups, (3) build a logistic regression risk model, (4) analyse attendance patterns, (5) generate a support priority ranking, and (6) simulate intervention scenarios.

Key findings:

- 33% of students are classified as At Risk ($G3 < 10$)
- Past failures (**failures**) is the strongest predictor of academic risk
- Students with 11+ absences have substantially higher at-risk rates

- Simulated full school support + high study time reduces risk for a meaningful proportion of currently at-risk students (see Section 6)

All findings are **associational**, not causal. This is observational data.

2 1. Dataset

Dataset: UCI Student Performance (Mathematics)
N = 394 students, 34 variables
At Risk (G3 < 10): 130 (33.0%)
Low Risk (G3 >= 10): 264 (67.0%)

3 2. Exploratory Data Analysis

Table 1: Descriptive statistics for key numeric variables

variable	n	mean	sd	median	q25	q75	min	max
G1	394	10.91	3.32	11	8	13	3	19
G2	394	10.72	3.77	11	9	13	0	19
G3	394	10.42	4.59	11	8	14	0	20
absences	394	5.71	8.01	4	0	8	0	75
studytime	394	2.04	0.84	2	1	2	1	4
failures	394	0.34	0.74	0	0	0	0	3

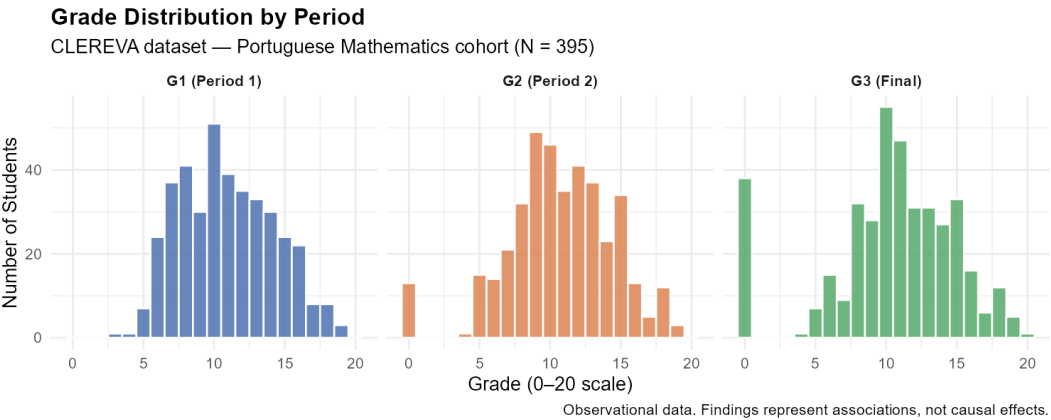


Figure 1: Grade distributions across three assessment periods

4 3. Group Comparisons

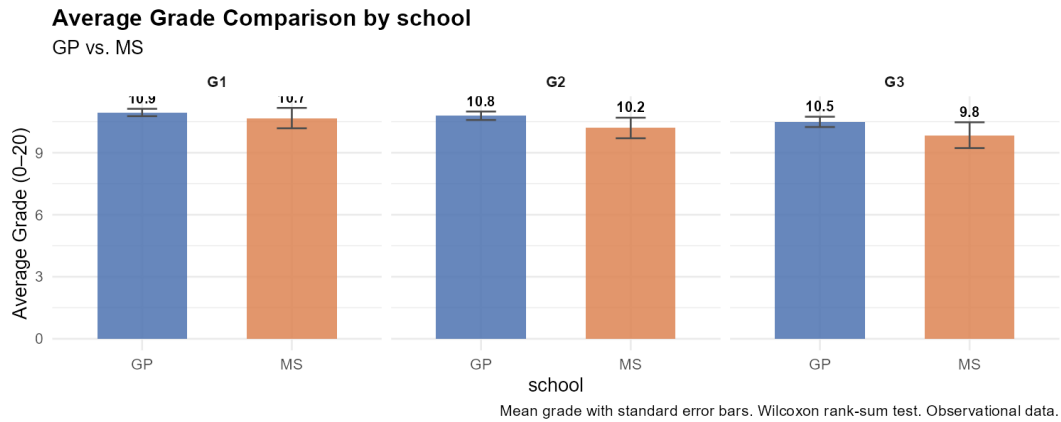


Figure 2: Average grade comparison: GP vs. MS students

Table 2: Wilcoxon test results: GP vs. MS (Bonferroni corrected)

Outcome	Group 1	Group 2	Median 1	Median 2	p (Bonf.)	Effect r
G1	GP	MS	11	10.5	1.0000	-0.053
G2	GP	MS	11	10.0	0.6045	-0.115
G3	GP	MS	11	10.0	0.5955	-0.116

5 4. Academic Risk Model

Table 3: Test set performance metrics

Metric	Value
Accuracy	0.3291
Sensitivity	0.6923
Specificity	0.1509
PPV	0.2857
NPV	0.5000
AUC_ROC	0.7025
N_total	394.0000
N_train	315.0000
N_test	79.0000
N_at_risk_train	104.0000
N_at_risk_test	26.0000

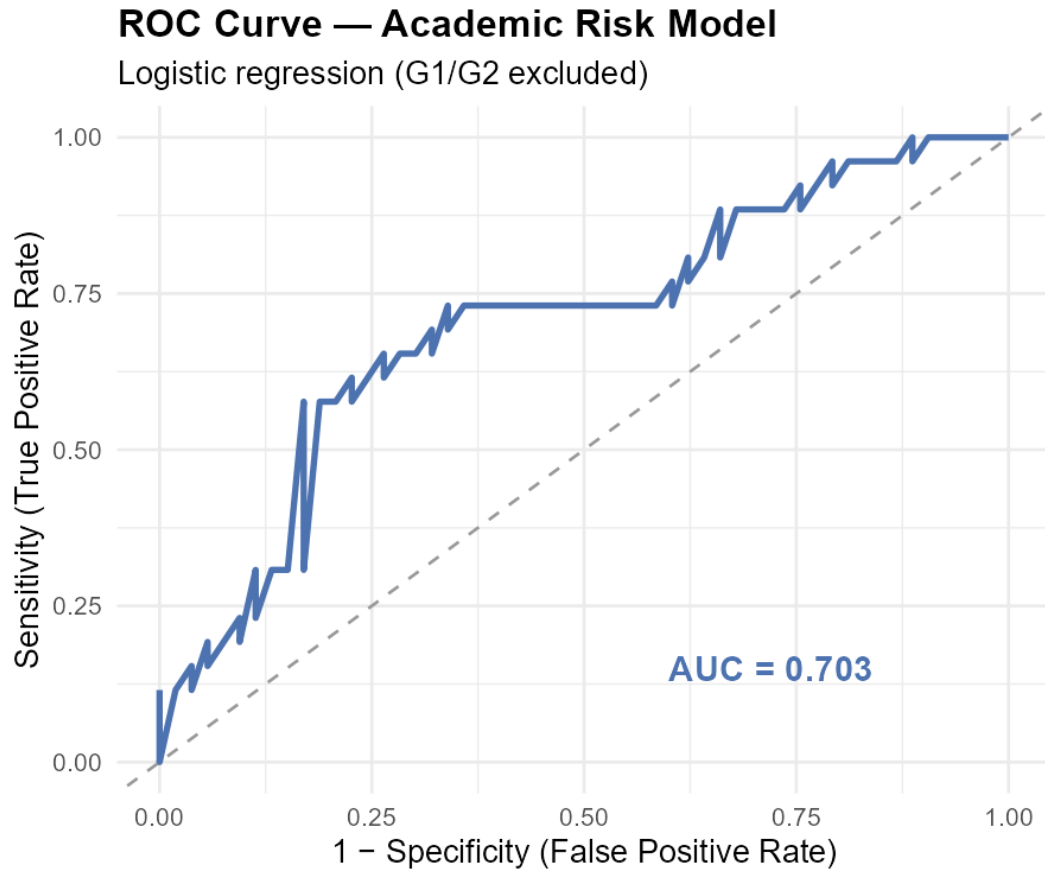


Figure 3: ROC curve — logistic regression risk model

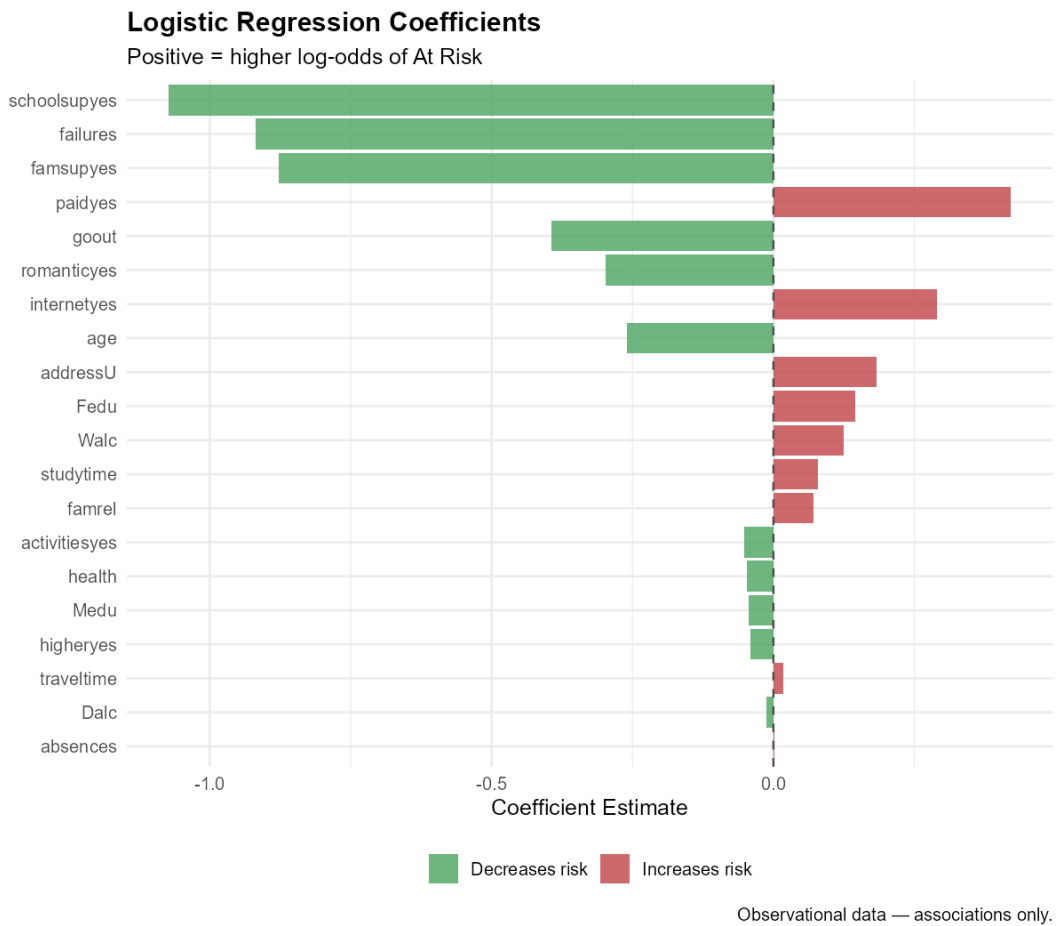


Figure 4: Logistic regression coefficients by predictor

6 5. Absence Analysis

Table 4: Grade outcomes and at-risk rates by absence tier

Tier	N	% Total	Median G3	Mean G3	% At Risk
0 absences	115	29.2	10	8.44	43.5
1–5	134	34.0	11	11.65	21.6
6–10	79	20.1	11	11.42	26.6
11+	66	16.8	10	10.15	45.5

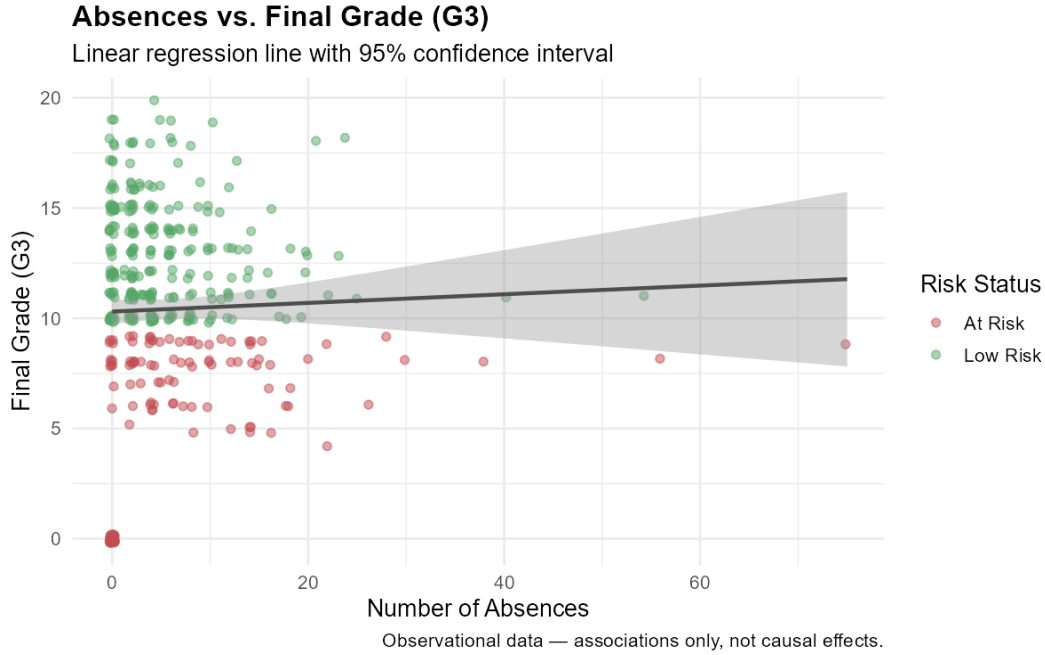


Figure 5: Absences vs. final grade (G3) with regression line

7 6. Scenario Analysis

Table 5: Summary of simulated intervention scenarios

Scenario	↓ Risk Prob.	At Risk→Low Risk	Mean Δ Prob.
Increased Study + School Support	343	88	-0.1480
Zero Absences	15	1	-0.0026

Scenario results are model extrapolations — not causal predictions.

8 7. Limitations

1. Observational data — no causal inference possible
2. Single cohort (N = 395) from two Portuguese schools, 2005–2006
3. Self-reported predictors subject to response bias
4. Binary risk threshold ($G3 < 10$) is a policy choice, not a natural boundary
5. Model evaluated on same-cohort holdout, not prospective validation

9 8. Responsible Use

This project generates decision-support indicators for counsellors. All outputs require human review before action. No personal identifiers are stored or displayed. Students are represented by anonymised indices only.

Generated on 30 July 2026 using CLEREVA v1.0.